2.9.1 Exponenciální funkce - příklady na oživení mocnin

Př. 1: Vypočti: a) 2^4 b) 3^2 c) 2^{-1} d) 3^{-2} e) $\left(\sqrt{2}\right)^4$

Př. 2: Vypočti: a) $\left(\frac{1}{2}\right)^2$ b) $\left(\frac{3}{5}\right)^2$ c) $\left(\frac{1}{4}\right)^{-2}$ d) $\left(\frac{2}{3}\right)^{-3}$ e) $8\cdot\left(\frac{2}{3}\right)^{-2}$

Př. 3: Vypočti: a) $\left(\sqrt{3}\right)^3$ b) $\left(\sqrt{8}\right)^3$ c) $\left(\frac{1}{\sqrt{3}}\right)^{-4}$ d) $\left(2\sqrt[3]{4}\right)^2$ e) $\left(\sqrt[3]{4}\right)^{-2}$

Př. 4: Vypočti: a) $2^2\cdot 4$ b) $\left(2\sqrt{2}\right)^{\frac{2}{3}}$ c) $4^{\frac{3}{2}}\cdot 2$ d) $\left(2^2\cdot 3\right)^2$

Př. 5: Zapiš jako jednu mocninu: a) $2^2\cdot 2^{x+1}$ b) $3^x\cdot 3^{1-x}$ c) $2^{x-2}\cdot 4^2$
d) $2^{x+2}\cdot\left(\frac{1}{2}\right)^{1-x}$ e) $\frac{2^2\cdot 2^{2x+1}}{2^{1-x}}$ f) $4^{x+1}\cdot 2^{2-x}$

Řešení na další stránce.

Př. 1: Vypočti: a) 2^4 b) 3^2 c) 2^{-1} d) 3^{-2} e) $\left(\sqrt{2}\right)^4$

a) $2^4 = 16$ b) $3^2 = 9$ c) $2^{-1} = \frac{1}{2}$

d) $3^{-2} = \frac{1}{3^2} = \frac{1}{9}$ e) $\left(\sqrt{2}\right)^4 = \left[\left(\sqrt{2}\right)^2\right]^2 = 2^2 = 4$

Př. 2: Vypočti: a) $\left(\frac{1}{2}\right)^2$ b) $\left(\frac{3}{5}\right)^2$ c) $\left(\frac{1}{4}\right)^{-2}$ d) $\left(\frac{2}{3}\right)^{-3}$ e) $8\cdot\left(\frac{2}{3}\right)^{-2}$

a) $\left(\frac{1}{2}\right)^2 = \frac{1^2}{2^2} = \frac{1}{4}$ b) $\left(\frac{3}{5}\right)^2 = \frac{9}{25}$ c) $\left(\frac{1}{4}\right)^{-2} = \frac{1}{\left(\frac{1}{4}\right)^2} = 4^2 = 16$

d) $\left(\frac{2}{3}\right)^{-3} = \left(\frac{3}{2}\right)^3 = \frac{27}{8}$ e) $8\cdot\left(\frac{2}{3}\right)^{-2} = 2^3 \cdot \frac{3^2}{2^2} = 2\cdot 9 = 18$

Př. 3: Vypočti: a) $\left(\sqrt{3}\right)^3$ b) $\left(\sqrt{8}\right)^3$ c) $\left(\frac{1}{\sqrt{3}}\right)^{-4}$ d) $\left(2\sqrt[3]{4}\right)^2$ e) $\left(\sqrt[3]{4}\right)^{-2}$

a) $\left(\sqrt{3}\right)^3 = 3^{\frac{3}{2}} = 3\cdot 3^{\frac{1}{2}} = 3\sqrt{3}$ b) $\left(\sqrt{8}\right)^3 = \left(\sqrt{2^3}\right)^3 = \left(2^{\frac{3}{2}}\right)^3 = 2^{\frac{9}{2}} = 2^{\frac{8}{2}}\cdot 2^{\frac{1}{2}} = 2^4\sqrt{2} = 16\sqrt{2}$

c) $\left(\frac{1}{\sqrt{3}}\right)^{-4} = \left(\sqrt{3}\right)^4 = 3^{\frac{4}{2}} = 3^2 = 9$ d) $\left(2\sqrt[3]{4}\right)^2 = 2^2\cdot\left(\sqrt[3]{2^2}\right)^2 = 2^2\cdot 2^{\frac{4}{3}} = 4\cdot 2\sqrt[3]{2} = 8\sqrt[3]{2}$

e) $\left(\sqrt[3]{4}\right)^{-2} = \frac{1}{\left(\sqrt[3]{4}\right)^2} = \frac{\sqrt[3]{4}}{\left(\sqrt[3]{4}\right)^2\cdot\sqrt[3]{4}} = \frac{\sqrt[3]{4}}{\left(\sqrt[3]{4}\right)^3} = \frac{\sqrt[3]{4}}{4}$

Př. 4: Vypočti: a) $2^2\cdot 4$ b) $\left(2\sqrt{2}\right)^{\frac{2}{3}}$ c) $4^{\frac{3}{2}}\cdot 2$ d) $\left(2^2\cdot 3\right)^2$

a) $2^2\cdot 4 = 4\cdot 4 = 16$ b) $\left(2\sqrt{2}\right)^{\frac{2}{3}} = \left(2\cdot 2^{\frac{1}{2}}\right)^{\frac{2}{3}} = \left(2^{1+\frac{1}{2}}\right)^{\frac{2}{3}} = \left(2^{\frac{3}{2}}\right)^{\frac{2}{3}} = 2^{\frac{3}{2}\cdot\frac{2}{3}} = 2^1 = 2$

c) $4^{\frac{3}{2}}\cdot 2 = \left(2^2\right)^{\frac{3}{2}}\cdot 2^1 = 2^3\cdot 2^1 = 2^4 = 16$ d) $\left(2^2\cdot 3\right)^2 = 2^4\cdot 3^2 = 16\cdot 9 = 144$

Př. 5: Zapiš jako jednu mocninu: a) $2^2\cdot 2^{x+1}$ b) $3^x\cdot 3^{1-x}$ c) $2^{x-2}\cdot 4^2$

d) $2^{x+2}\cdot\left(\frac{1}{2}\right)^{1-x}$ e) $\frac{2^2\cdot 2^{2x+1}}{2^{1-x}}$ f) $4^{x+1}\cdot 2^{2-x}$

a) $2^2\cdot 2^{x+1} = 2^{2+x+1} = 2^{x+3}$ b) $3^x\cdot 3^{1-x} = 3^{x+1-x} = 3^1 = 3$

c) $2^{x-2} \cdot 4^2 = 2^{x-2} \cdot \left(2^2\right)^2 = 2^{x-2} \cdot 2^4 = 2^{x-2+4} = 2^{x+2}$

d) $2^{x+2} \cdot \left(\frac{1}{2}\right)^{1-x} = 2^{x+2} \cdot \left(2^{-1}\right)^{1-x} = 2^{x+2} \cdot 2^{x-1} = 2^{2x+1}$

e) $\frac{2^2 \cdot 2^{2x+1}}{2^{1-x}} = 2^{2+2x+1-(1-x)} = 2^{2x+3-1+x} = 2^{3x+2}$

f) $4^{x+1} \cdot 2^{2-x} = \left(2^2\right)^{x+1} \cdot 2^{2-x} = 2^{2x+2} \cdot 2^{2-x} = 2^{x+4}$